Lecture 7: Circuit Graphs and the Incidence Matrix

EE201 – Circuit Theory I

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current voltage nodes a path in a graph

Interconnection equations

A circuit with b branches: $2b$ unknowns, v_k and i_k for each branch k.

Group	How many	Describes
element equations	b	*what* each element is: $v = Ri$, a source value, ...
interconnection equations	b	*how* they are wired: $n-1$ KCL, $b-n+1$ KVL

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KCL and KVL do not care what the elements are

Replace a resistor by a capacitor or a source, keep the wiring: KCL and KVL stay the same. Keep only the wiring — the graph — and write all KCL equations as one matrix equation.

The graph of a circuit

Circuit graph

Replace every two-terminal element by a line segment, a branch. The points where branches meet are the nodes. The graph records only *which nodes each branch joins*; the shape of the drawing carries no information.

Procedure

1. Label each branch: $1, \ldots, b$.
2. Give each branch a direction — arbitrary.
3. Label each node: $\textcircled{1}$, $\ldots$, $\textcircled{n}$.
4. Draw the graph: dots and arrows.

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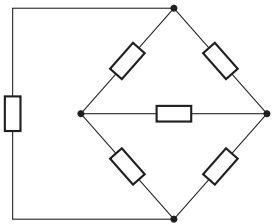
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The arrow is the reference direction of i_k. With associated references, v_k has $+$ at the tail.

If the current flows the other way, i_k is simply negative.

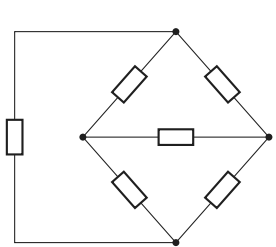
Example: the graph of a bridge circuit



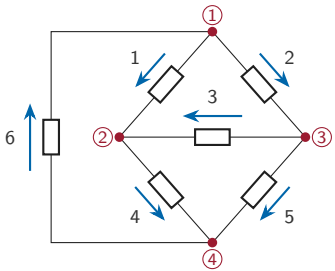
circuit $\mathcal{N}$

Six elements, drawn as boxes: their type does not matter here.

Example: the graph of a bridge circuit



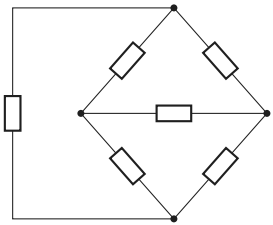
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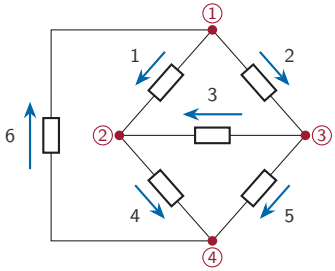
steps 1–3

Number the branches, give each an arrow (current reference), number the nodes.

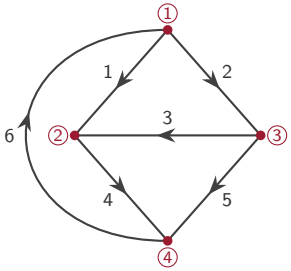
Example: the graph of a bridge circuit



circuit $\mathcal{N}$



steps 1–3



step 4: the graph

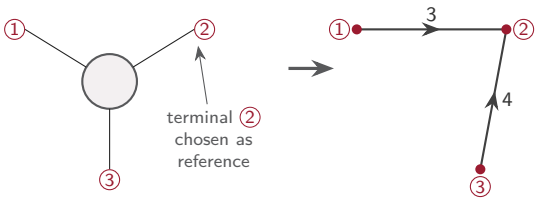
Nodes $\{①, ②, ③, ④\}$, $n = 4$; branches $\{1, \ldots, 6\}$, $b = 6$.

Branch 6 is curved only for clarity — a straight arrow is the same graph.

Elements with more than two terminals

A k-terminal element

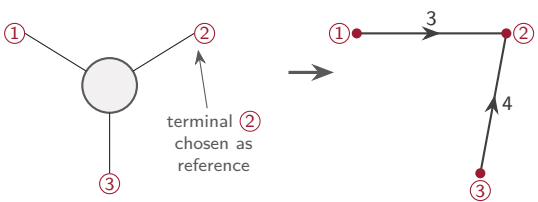
Choose one terminal as its reference (arbitrary). Draw $k-1$ branches, one from each other terminal to the reference terminal.



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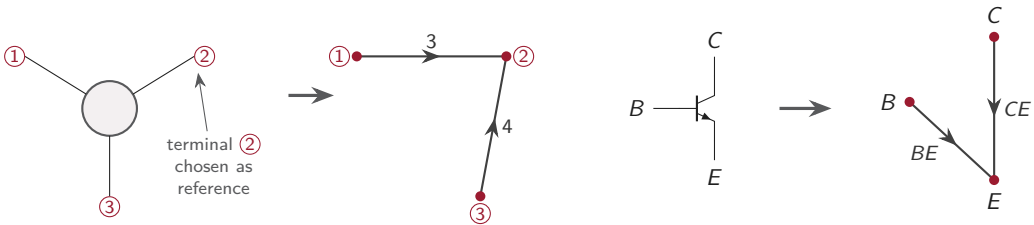


Why $k-1$? KCL on a surface around the element: the k terminal currents sum to zero, so only $k-1$ are free. By KVL, the $k-1$ voltages to the reference fix all the others.

Elements with more than two terminals

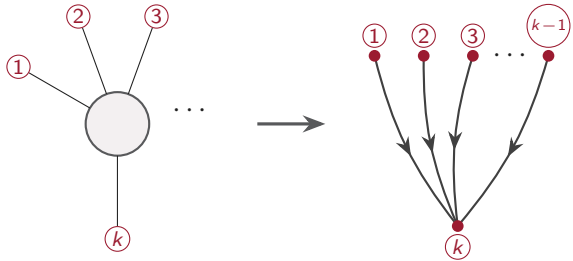
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Why $k-1$? KCL on a surface around the element: the k terminal currents sum to zero, so only $k-1$ are free. By KVL, the $k-1$ voltages to the reference fix all the others. A transistor: emitter as reference, two branches.

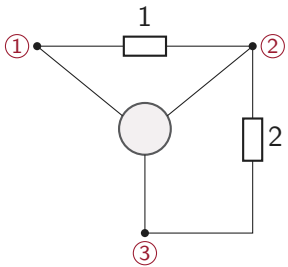
The general case



a k-terminal element $k-1$ **branches, all to terminal** k

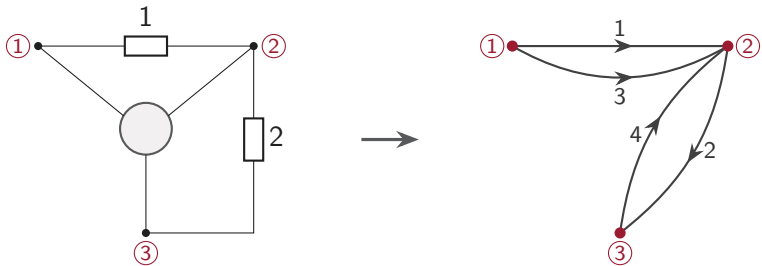
A dependent source needs nothing new: it is already two branches — the branch that senses the controlling variable, and the source branch.

Example: a circuit with a 3-terminal element



Elements 1 and 2 have two terminals; the circle has three. Find the graph.

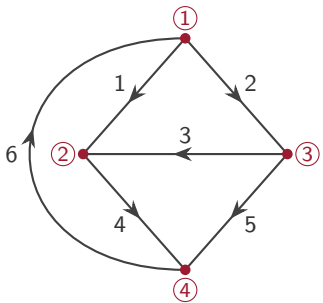
Example: a circuit with a 3-terminal element



Elements 1, 2: one branch each. 3-terminal element, reference terminal at ②: branches 3 (①→②) and 4 (③→②).

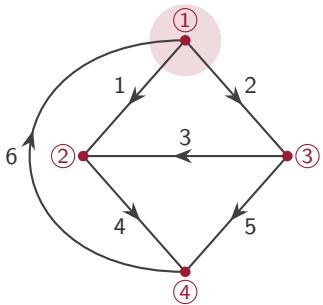
$n = 3$, $b = 4$. Two branches may join the same pair of nodes.

KCL on a circuit graph



Current along its arrow. At each node:
leaving $+$, entering $-$, sum $= 0$.

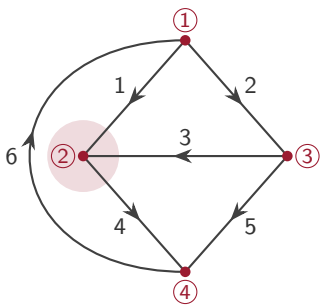
KCL on a circuit graph



Current along its arrow. At each node:
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$$\textcircled{1}: \quad i_1 + i_2 - i_6 = 0$$

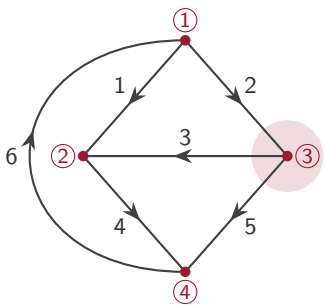
KCL on a circuit graph



Current along its arrow. At each node:
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$$\begin{aligned} ①: \quad & i_1 + i_2 - i_6 = 0 \\ ②: \quad & -i_1 - i_3 + i_4 = 0 \end{aligned}$$

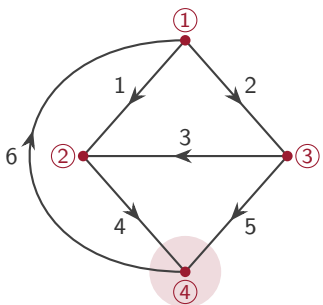
KCL on a circuit graph



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\end{aligned}
$$

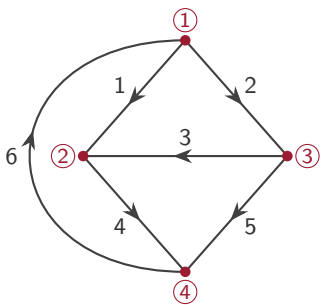
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$$

Only the graph was used. Same graph $\Rightarrow$ same KCL equations, whatever the elements are.

From equations to a matrix

Only $+1$, -1 and 0 appear. Rows: nodes. Columns: branches.

$$\begin{array}{c} \\ ① \\ ② \\ ③ \\ ④ \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 1 \end{bmatrix} \end{array} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \\ i_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \Longleftrightarrow \quad A_a \mathbf{i} = \mathbf{0}$$

A matrix that stores the graph: a circuit matrix. This one is the first and most basic.

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$$
\begin{array}{c}
 \\
① \\
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$$

A row is a node: row ① times $\mathbf{i}$ is KCL at ①.

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A column is a branch: one $+1$ (leaves) and one -1 (enters). Column 3: from ③ to ②.

The incidence matrix

Incidence matrix A_a $(n \times b)$

$$a_{ij} = \begin{cases} +1 & \text{if branch } j \text{ leaves node } i, \\ -1 & \text{if branch } j \text{ enters node } i, \\ 0 & \text{otherwise.} \end{cases}$$

With $\mathbf{i} = [\, i_1 \ \cdots \ i_b \,]^T$, the branch current vector: KCL at every node is $A_a\mathbf{i} = \mathbf{0}$.

- The subscript a: *all* nodes have a row.

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- Reversing an arrow flips the sign of that column — nothing else changes.

Review: linear dependence

Linear dependence

$\mathbf{v}_1, \ldots, \mathbf{v}_k$ (same size) are linearly dependent if there are real $\alpha_1, \ldots, \alpha_k$, **not all zero**, with $\alpha_1 \mathbf{v}_1 + \cdots + \alpha_k \mathbf{v}_k = \mathbf{0}$. Otherwise they are linearly independent.

Review: linear dependence

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$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

independent: $[\alpha_1 \quad -\alpha_2]^T = \mathbf{0}$ only if both are zero.

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$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$$

dependent: $\mathbf{v}_1 - 2\mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$, so $\mathbf{v}_3 = 2\mathbf{v}_2 - \mathbf{v}_1$.

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Rows of M dependent $\Rightarrow$ one equation of $M\mathbf{x} = \mathbf{0}$ follows from the others.
Number of independent rows: the rank. All rows independent: full row rank.

The rows of A_a are always dependent

Every column has one $+1$ and one -1, so

$$\text{row } \textcircled{1} + \text{row } \textcircled{2} + \cdots + \text{row } \textcircled{n} = [0\ 0\ \cdots\ 0]$$

— dependence with every $\alpha = 1$. One KCL equation is always redundant.

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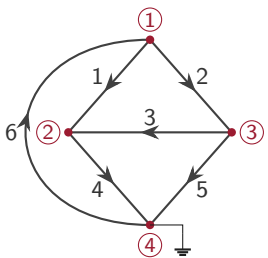
Reduced incidence matrix A $((n-1) \times b)$

Choose a reference node (ground), arbitrarily. Delete its row from A_a.

The reference node is also the common reference for node voltages.

Example: the reduced incidence matrix of $\mathcal{N}$

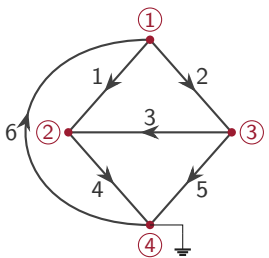
Reference node: ④.



$$A_a = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 1 \end{bmatrix}$$

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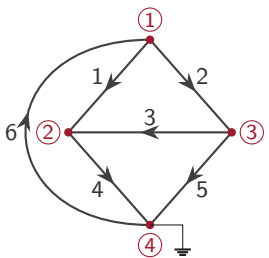
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$$A = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \end{bmatrix} \qquad (3 \times 6)$$

Example: the reduced incidence matrix of $\mathcal{N}$



Reference node: ④.

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$$A = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \end{bmatrix} \qquad (3 \times 6)$$

Rows independent: columns 6, 4, 5 each have a single nonzero entry, in rows ①, ②, ③.

Deleting the row loses nothing

No equation is lost

$A\mathbf{i} = \mathbf{0} \Rightarrow A_a\mathbf{i} = \mathbf{0}$: the deleted row is minus the sum of the rows that were kept.

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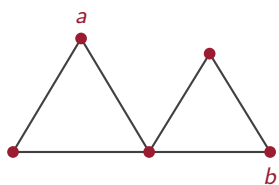
No part of the graph is lost

Each column of A_a sums to zero, so the deleted row can be rebuilt from A.
A column of A with a single nonzero entry is a branch that touches the reference node.

Connected graphs

Connected graph

Any two nodes are joined by a path of branches, ignoring the arrow directions.



connected

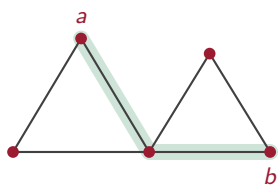


not connected: two separate pieces

Connected graphs

Connected graph

Any two nodes are joined by a path of branches, ignoring the arrow directions.



a path
from a to b

connected



not connected: two separate pieces

Connected graph $\Rightarrow$ rows of A independent

Full row rank

If the graph is connected, the $n-1$ rows of A are linearly independent: $\operatorname{rank} A = n-1$.

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Full row rank

If the graph is connected, the $n-1$ rows of A are linearly independent: $\operatorname{rank} A = n-1$.

Why. Suppose $\sum \alpha\,(\text{rows of } A) = 0$; the reference node has no row, give it $\alpha = 0$.

- A branch from p to q gives $\alpha_p - \alpha_q = 0$: neighbours have equal α.
- Connected: every node has a path to the reference, so every $\alpha = 0$.

Connected graph $\Rightarrow$ rows of A independent

Full row rank

If the graph is connected, the $n-1$ rows of A are linearly independent: $\operatorname{rank} A = n-1$.

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- A branch from p to q gives $\alpha_p - \alpha_q = 0$: neighbours have equal α.
- Connected: every node has a path to the reference, so every $\alpha = 0$.

Two separate pieces (for example, coupled only through a transformer): the other piece keeps its own dependence. Join the pieces with one wire — by KCL it carries no current, so nothing changes. In EE201 all graphs are connected.

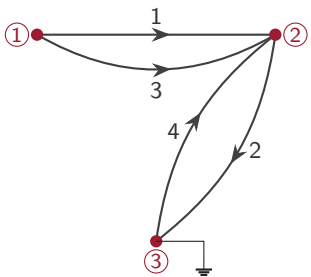
KCL of a circuit graph

$$A\,\mathbf{i} = \mathbf{0}$$

$A \in \mathbb{R}^{(n-1)\times b}$	reduced incidence matrix
$\mathbf{i} \in \mathbb{R}^{b}$	branch current vector

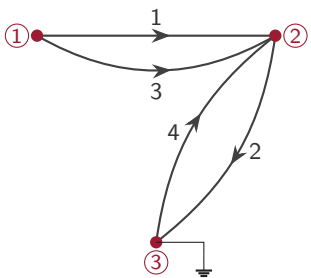
For a connected graph: $n-1$ independent KCL equations,
and every other KCL equation of the circuit follows from them.

Example: KCL for the circuit with a 3-terminal element



Reference node: ③.

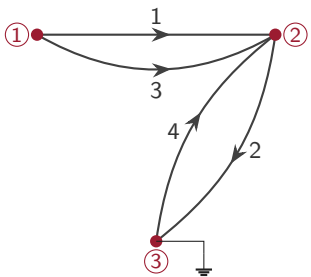
Example: KCL for the circuit with a 3-terminal element



$$A_a = \begin{array}{c} \\ ① \\ ② \\ ③ \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 1 & -1 & -1 \\ 0 & -1 & 0 & 1 \end{bmatrix} \end{array}$$

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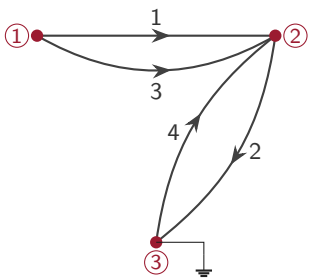


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$$A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 1 & -1 & -1 \end{bmatrix}$$

$$i_1 + i_3 = 0, \qquad -i_1 + i_2 - i_3 - i_4 = 0$$

Columns 1 and 3 are equal: same nodes, same direction. Reference terminal (②) and reference node (③): two separate choices.

Summary

- $2b$ equations: b **element** equations and b **interconnection** equations (KCL, KVL). The second group depends only on the graph.
- The graph: number the branches, give each an arbitrary arrow, number the nodes. A k-terminal element becomes $k-1$ branches.
- **Incidence matrix** A_a ($n \times b$): $+1$ leaves, -1 enters. KCL: $A_a\mathbf{i} = \mathbf{0}$. Its rows always sum to zero.
- **Reduced incidence matrix** A: delete the reference node's row. For a connected graph, $A\mathbf{i} = \mathbf{0}$ gives exactly the $n-1$ independent KCL equations.

Next lecture

KVL on the graph

Every branch voltage is a difference of node voltages: $\mathbf{v} = A^T\mathbf{e}$ — the same matrix A, transposed. Then Tellegen's theorem, in one line from $A\mathbf{i} = \mathbf{0}$ and $\mathbf{v} = A^T\mathbf{e}$.

Questions?